

Name: Shweta Chauhan

Lab: Lab 2

Date: 3/20/2022

a. Program Description:

The purpose of this program is to calculate car efficiency based on inputs from the user for miles per gallon, capacity and cost per gallon. Once the car efficiency is calculated, the program display how efficient is the car. To get inputs from the suer I have used input() function. Based on the inputs, cost of driving 100 miles and distance on a tank of gas is calculated.

After that if, elif and else statements are used to examine how efficient the car is based on miles per gallon. In order to do that comparison operators are used within if else statements to reach the results.

b. Program Source Code:

```
#This program is to calculate efficiency of a car based on inputs from the user
name = "CNET-142 - Shweta Chauhan"
#Printing Name based on the variable defined as name
print("{:14}".format("Name"), ":", name)
lab = "Lab 2 - Car Mileage"
#Printing Lab name based on the variable defined as lab
print("{:14}".format("Lab"), ":", lab)
#Taking current date and time
currentTime = datetime.now()
print(currentTime)
timestamp_str = currentTime.strftime("%b-%d-%Y %a (%I:%M:%S%p)")
#Printing current date and time based on the variables defined as timestamp_str
print("{:14}".format("Current Time"), ":", timestamp_str)

capacity = float(input("Enter the capacity of the car's gas tank (in gallons): "))
miles_per_gallon = int(input("Enter car's miles per gallon: "))
price_per_gallon = float(input("Enter price per gallon: "))
print("Cost for driving 100 miles is : $ %.2f" % ((100/miles_per_gallon)*price_per_gallon))
print("Distance on a tank of gas is: %.2f miles" % (capacity*miles_per_gallon))
if miles_per_gallon < 30:
    print("Your car MPG is %.1f . It is not fuel efficient car" %miles_per_gallon )
elif 30 < miles_per_gallon < 40:
    print("Your car MPG is %.1f . It is average fuel efficient car." %miles_per_gallon )
elif 40 < miles_per_gallon < 50:
    print("Your car MPG is %.1f . It is fuel efficient car" %miles_per_gallon)
else:
    print("Your car MPG is %.1f . It is very fuel efficient car" %miles_per_gallon)
```

c. Program Output

Test Case #1

```
C:\Users\shwetachauhan\PycharmProjects\pythonProject\venv\Scripts\py
Name           : CNET-142 - Shweta Chauhan
Lab            : Lab 2 - Car Efficiency
2022-03-20 23:43:36.924623
Current Time   : Mar-20-2022 Sun (11:43:36PM)
Enter the capacity of the car's gas tank (in gallons): 18.5
Enter car's miles per gallon: 33
Enter price per gallon: 3.89
Cost for driving 100 miles is : $ 11.79
Distance on a tank of gas is: 610.50 miles
Your car MPG is 33.0 . It is average fuel efficient car.
```

Test Case #2

```
C:\Users\shwetachauhan\PycharmProjects\pythonProject\venv\Scripts\python.exe C:
Name           : CNET-142 - Shweta Chauhan
Lab            : Lab 2 - Car Mileage
2022-03-20 23:46:25.639873
Current Time   : Mar-20-2022 Sun (11:46:25PM)
Enter the capacity of the car's gas tank (in gallons): 18.5
Enter car's miles per gallon: 45
Enter price per gallon: 3.89
Cost for driving 100 miles is : $ 8.64
Distance on a tank of gas is: 832.50 miles
Your car MPG is 45.0 . It is fuel efficient car
```

Test Case #3

```
C:\Users\shwetachauhan\PycharmProjects\pythonProject\venv\Scripts\py
Name           : CNET-142 - Shweta Chauhan
Lab            : Lab 2 - Car Mileage
2022-03-20 23:47:07.223053
Current Time    : Mar-20-2022 Sun (11:47:07PM)
Enter the capacity of the car's gas tank (in gallons): 18.5
Enter car's miles per gallon: 60
Enter price per gallon: 3.89
Cost for driving 100 miles is : $ 6.48
Distance on a tank of gas is: 1110.00 miles
Your car MPG is 60.0 . It is very fuel efficient car
```